Deliverable #1 - Game Proposal

ITCS-5236-001: AI in Gaming
Project Group 22
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Game Title

Echo Heist: Adaptive Multi-Agent Stealth Game

Game Description

Genre: 2D Top-Down Stealth / Strategy
Theme: Futuristic security facility / heist

Echo Heist is a 2D top-down stealth game where the player must infiltrate a secure facility, steal a protected artifact and escape without being caught by AI-controlled security guards.

The main focus of the game is intelligent enemy behavior. Rather than simply following the player, guards can detect suspicious activity, communicate with each other, search areas strategically, protect important locations and adapt to the player's behavior. The goal is to make each attempt feel different as the guards respond to how the player plays.

The game will be developed in Unity with a small facility map and short replayable rounds.

Game Mechanics

Major Mechanics

- **Player Movement:** The player moves freely through the 2D facility while avoiding security guards.
- **Stealth and Detection:** Guards have a limited field of view. The player must avoid being seen or use the environment to break line of sight.
- **Objective and Escape:** The player must reach the artifact, steal it and successfully reach an escape point.
- **Guard States:** Guards can patrol, investigate suspicious activity, search for the player, chase the player, guard important locations and return to patrol.

- **Coordinated Guard Behavior:** Guards communicate when the player is detected and can take different roles instead of all directly chasing the player.

Minor Mechanics

- Sound or distraction events can temporarily attract guards.
- Guards use pathfinding to move around walls and obstacles.
- A spatial heat map represents areas where the player is likely to be after escaping detection.
- The game tracks player behavior across rounds, such as commonly used routes or escape points.
- Difficulty can adjust based on player performance by changing how effectively guards search and coordinate.

Main AI Component

The main AI component is an **adaptive multi-agent guard system**. Each security guard acts as an individual AI agent, while all guards also share information and coordinate as a team.

Each guard uses a **Finite State Machine** to control behaviors such as Patrol, Investigate, Search, Chase, Intercept and Guard. Guards use field-of-view calculations and line-of-sight checks to determine whether they can see the player. **A* pathfinding** is used to navigate through the facility.

When the player is detected, information such as the player's last known position is stored in a **shared memory system**. Guards use this information together with a **spatial heat map** that estimates where the player may have moved after leaving sight. A utility-based decision system determines which action is most useful for each guard. For example, one guard may chase the player while another searches a nearby room and another protects the escape route.

The game will also maintain a small **long-term memory of important events from previous rounds**, such as frequently used escape routes, successful distractions and previous guard strategies. A higher-level AI planning component can use these past events to adjust team strategy in later rounds. For example, if the player repeatedly escapes through the same route, a guard may be assigned to protect that location during a future attempt.

This combination of **perception, FSM behavior, A* pathfinding, tactical spatial analysis, utility-based decisions, shared memory, agent cooperation and adaptive behavior** forms the primary complex AI system of the game.